

Tommy Nicol

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PROFESSIONAL SUMMARY

Computer Science student with a Philosophy minor building AI agent workflows and client software. Combines practical AI development with the ability to explain AI capabilities to nontechnical users and apply them to users' goals, informed by ongoing study of bias, alignment, and human agency.

EDUCATION

University of Pittsburgh

Expected December 2027

B.S. in Computer Science; Minor in Philosophy

GPA: 3.3

Coursework in Progress: Machine Learning; AI Architectures; Detecting Bias and Hallucinations in AI Systems; Modern Philosophical Problems with AI.

Additional Coursework: History of AI; Data Structures and Algorithms; Low-Level Computing; Logic; Spanish language coursework.

TECHNICAL EXPERIENCE

Freelance Web Developer

July 2026–Present

- Built two paid client websites for a student and a local bar; translated client goals into scoped requirements, site content, functionality, and launch criteria.
- Developed an agent-assisted workflow for planning, implementation, review, and browser testing, enabling faster prototype production to support outreach to prospective clients and help clients compare ideas and refine requirements.

SELECTED AI PROJECTS & RESEARCH

Adaptive Development Orchestrator | *JavaScript, Claude Code, JSON Schema*

- Built an AI development framework that analyzes existing code, breaks requests into manageable tasks, and coordinates agents to complete them in the right order.
- Routed tasks by difficulty and model capability, assigning simpler work to smaller models to manage cost and reserve stronger models for complex reasoning.
- Configured separate review and verification agents to document acceptance criteria, test commands, exit codes, and pass/fail/unverified results; specified explicit user approval for releases.

GPT-Style Language Model | *Python, PyTorch*

- Implemented and trained a GPT-style transformer on publicly available text, including token embeddings, causal multi-head attention, residual connections, and layer normalization.
- Built a next-token prediction training loop and examined the model's generated text.

Text-to-Speech System | *Python, Tkinter, Claude API, Edge TTS*

- Built and released a macOS app that converts PDF, Word, and text documents into spoken audio, combining rule-based and Claude-assisted cleanup of citations, headers, and OCR artifacts.
- Added English/Spanish translation and narration, pitch-preserving playback speed controls, and saved transcripts to make dense readings accessible through audio.

AI Alignment & Human Agency | *Philosophy research in progress*

- Investigating how AI alignment should account for human agency, with the goal of informing the design of human-AI systems that improve efficiency while preserving people's ability to make decisions and direct their work.

TECHNICAL SKILLS

Programming: Python, Java, C, R, MIPS Assembly

AI & ML: Model selection based on capabilities, limitations, and cost; PyTorch, NumPy, transformers, agent orchestration, context engineering, Claude Code, Codex

Web & Tools: API integration, Cloudflare, GitHub Actions, Git, Linux/Unix, shell scripting

Spoken Languages: English (fluent), Spanish (conversational)

ADDITIONAL EXPERIENCE

High-Volume Hospitality — Restaurants and Bars

2021–Present

- Coordinated with coworkers and communicated with guests during high-volume restaurant and bar shifts; balanced 15+ hours of weekly work with full-time coursework.